

Scattering by a Central Potential - Approved Segments

11 approved segments, in reading order

scattering by a central potential

OCT 22 2024

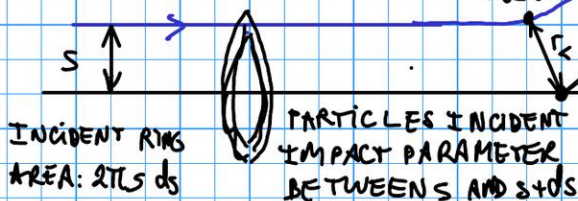
INCIDENT ENERGY $E = \frac{m}{2} v_0^2$

INCIDENT VELOCITY $v_0 = \sqrt{2E/m}$

ANGULAR MOMENTUM $l = S m v_0 = S \sqrt{2mE}$

$S = \frac{l}{\sqrt{2mE}}$

APSIDAL POINT



$S \leftrightarrow \theta$

PARTICLES EMERGING WITH SCATTERING ANGLE BETWEEN θ AND $\theta+d\theta$

$d^3r = r^2 dr d\phi \sin\theta d\theta$

SCATTERED RING SOLID ANGLE: $d\Omega = 2\pi \sin\theta d\theta$

$S(\theta, E)$

Azimuthal angle ϕ irrelevant by symmetry

INCIDENT INTENSITY

$I = \frac{\text{\# PART INCIDENT PER UNIT TIME}}{\text{INCIDENT BEAM CROSS SECTION AREA}}$

DIFFERENTIAL CROSS SECTION

$\sigma(r) d\Omega = \frac{\text{\# PART SCATTERED INTO } d\Omega \text{ PER UNIT TIME}}{\text{INCIDENT INTENSITY}}$

$I (2\pi S) |ds| = \text{\# part going through ring between } S \text{ and } S+ds = \text{\# part emerging with angle between } \theta \text{ and } \theta+d\theta = I \sigma(\theta) 2\pi \sin\theta |d\theta|$

THEREFORE:

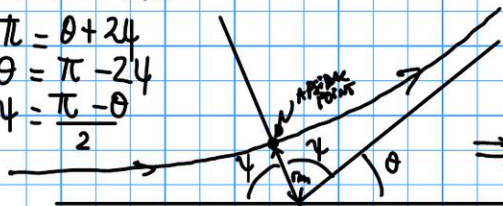
$\sigma(\theta) = \frac{S}{\sin\theta} \left| \frac{ds}{d\theta} \right|$

DEFINE ALSO THE TOTAL CROSS SECTION:

$\sigma_T = \int d\Omega \sigma(r)$

REPULSIVE CASE:

$$\begin{aligned}\pi &= \theta + 2\psi \\ \theta &= \pi - 2\psi \\ \psi &= \frac{\pi - \theta}{2}\end{aligned}$$

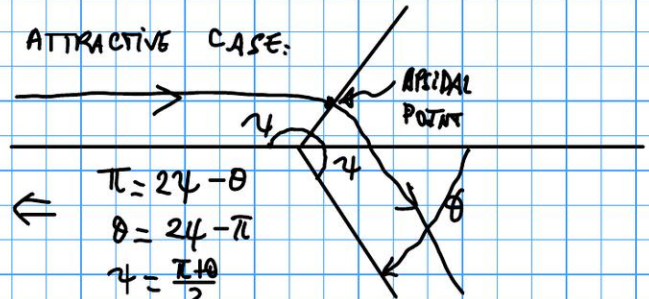


IN ALL CASES:

$$\theta = |\pi - 2\psi|$$

ATTRACTIVE CASE:

$$\begin{aligned}\pi &= 2\psi - \theta \\ \theta &= 2\psi - \pi \\ \psi &= \frac{\pi + \theta}{2}\end{aligned}$$



COMPUTING $S(\theta, E)$ (UNIVERSAL METHOD)

$$E = \frac{m}{2} \dot{r}^2 + V(r) + \frac{l^2}{2mr^2} \quad \dot{r} = \left[\frac{2}{m} \left(E - V(r) - \frac{l^2}{2mr^2} \right) \right]^{1/2} \quad \dot{r} = \frac{dr}{dt} \Rightarrow dt = \frac{dr}{\dot{r}}$$

$$l = mr^2 \dot{\theta} \Rightarrow \frac{d\theta}{dt} = \dot{\theta} = \frac{l}{mr^2} \Rightarrow d\theta = \dot{\theta} dt = \frac{l}{mr^2} dt = \frac{l}{mr^2} \frac{dr}{\dot{r}} = \frac{l}{mr^2} \frac{r^2 dr}{\left[\frac{2}{m} \left(E - V(r) - \frac{l^2}{2mr^2} \right) \right]^{1/2}} = \frac{l}{m} \frac{\sqrt{m}}{\sqrt{2E}} \frac{r^2 dr}{\left[1 - \frac{V(r)}{E} - \frac{l^2}{2mEr^2} \right]^{1/2}}$$

$$\psi = \int_0^\psi d\psi = \int_{\theta_{\min}}^\pi d\theta = \int_{r_{\min}}^\infty \frac{s r^{-2} dr}{\left[1 - V(r) - s^2/r^2 \right]^{1/2}} \Rightarrow \theta(\psi) = \left| \pi - 2 \int_0^{u_{\min}} \frac{s du}{\left[1 - \frac{V(u)}{E} - s^2 u^2 \right]^{1/2}} \right|$$

$$u = 1/r \quad du = -r^{-2} dr \quad u_{\min} = 1/r_{\min}$$

From here, compute $\left| \frac{ds}{d\theta} \right| = \frac{1}{\left| \frac{d\theta}{ds} \right|}$ and use it to get $\sigma(\theta) = \frac{S}{\sin \theta} \left| \frac{ds}{d\theta} \right|$

UP TO HERE, RESULTS ARE VALID FOR ANY CENTRAL POTENTIAL $V(r)$

Coulomb Scattering

$E > 0$	hyperbolic
$E = 0$	parabolic
$E < 0$	elliptic
ONLY FOR $V = -\frac{k}{r}$!!!	

FORCE $F = \frac{Z Z' e^2}{r^2}$

$f = -\frac{k}{r^2}$ $V = -\frac{k}{r}$
 $k = -Z Z' e^2$

Scattering: $E > 0$

$\frac{1}{r} = G [1 + E \cos(\tilde{\theta} - \tilde{\theta}')]]$

$G = \frac{mk}{l^2} = -\frac{m}{l^2} Z Z' e^2$

Eccentricity $E = \left[1 + \frac{2E l^2}{m k^2} \right]^{\frac{1}{2}} = \left[1 + \frac{2E l^2}{m (Z Z' e^2)^2} \right]^{\frac{1}{2}} = \left[1 + \left(\frac{2sE}{Z Z' e^2} \right)^2 \right]^{\frac{1}{2}} > 1$

$l = s m v_0 = s \sqrt{2mE} \rightarrow \frac{2 l^2 E}{m} = \frac{s^2 2^2 m E^2}{m} = (2sE)^2$

2 SITUATIONS: $\Rightarrow$ REPULSIVE CASE $Z Z' > 0 \Rightarrow k = -Z Z' e^2 < 0$

2 SITUATIONS: i) REPULSIVE CASE $zz' > 0 \Rightarrow k = -zz'e^z < 0$

ii) ATTRACTIVE CASE $zz' < 0 \Rightarrow k = -zz'e^z > 0$ (ALSO GRAVITY $k = Gm_1m_2$)

i) REPULSIVE CASE: $k < 0$ $Q = \frac{mk}{l^2} < 0$

$$\frac{1}{r} = Q[1 + \epsilon \cos(\tilde{\theta} - \tilde{\theta}')] > 0 \Leftrightarrow 1 + \epsilon \cos(\tilde{\theta} - \tilde{\theta}') < 0 \Leftrightarrow \epsilon \cos(\tilde{\theta} - \tilde{\theta}') < -1 \Leftrightarrow \cos(\tilde{\theta} - \tilde{\theta}') < -\frac{1}{\epsilon}$$

We need the cosine to be negative and larger in magnitude than $1/\epsilon$

In this case, $\tilde{\theta} = \tilde{\theta}'$ is not a physical point, because $\cos(\tilde{\theta} - \tilde{\theta}') = \cos(0) = 1 \Rightarrow \frac{1}{r} = Q[1 + \epsilon] < 0$

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UNPHYSICAL

SO $\tilde{\theta} = \tilde{\theta}'$ IS IN THE MIDDLE OF THE INACCESSIBLE REGION OF THE HYPERBOLA, AND $\tilde{\theta} = \tilde{\theta}' \pm \pi$ WILL BE THE PERIAPSIS, SO WE CHOOSE $\tilde{\theta}' = \pi$ AND WE GET

$$\cos(\tilde{\theta} - \tilde{\theta}') = \cos(\tilde{\theta} - \pi) = \cos \tilde{\theta} \underbrace{\cos \pi}_{-1} + \sin \tilde{\theta} \underbrace{\sin \pi}_0 = -\cos \tilde{\theta}$$

$$\frac{1}{r} = \left(-\frac{mk}{l^2}\right) [-1 - \epsilon \cos(\tilde{\theta} - \tilde{\theta}')] = \frac{mz_1 z_2 e^2}{l^2} [\epsilon \cos \tilde{\theta} - 1]$$

MAXIMUM AT $\tilde{\theta} = 0 \Rightarrow$

$$\text{MINIMUM } r: \frac{1}{r_m} = \frac{mz_1 z_2 e^2}{l^2} [\epsilon - 1]$$

SO THE ANGLE AT THE APSIDAL POINT IS $\tilde{\theta} = 0$

$$\frac{1}{r} = \left(-\frac{mk}{l^2}\right) [-1 - \epsilon \cos(\tilde{\theta} - \tilde{\theta}')] = \frac{mZz'e^2}{l^2} [\epsilon \cos \tilde{\theta} - 1] \quad \text{MINIMUM } r: \frac{1}{r_m} = \frac{mZz'e^2}{l^2} [\epsilon - 1]$$

SO THE ANGLE AT THE APSIDAL POINT IS $\tilde{\theta} = 0$

WE NOW NEED $\psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}(\frac{1}{r} \rightarrow 0)$

[IN THE REPULSIVE CASE]
 $\theta = \pi - 2\psi \Rightarrow \psi = \frac{\pi - \theta}{2}$

$$0 = \epsilon \cos \psi - 1 \Leftrightarrow \frac{1}{\epsilon} = \cos \psi = \cos\left(\frac{\pi - \theta}{2}\right) = \underbrace{\cos \frac{\pi}{2}}_0 \cos \frac{\theta}{2} + \underbrace{\sin \frac{\pi}{2}}_1 \sin \frac{\theta}{2} = \sin \frac{\theta}{2}$$

ii) ATTRACTIVE CASE: $k > 0 \quad C = \frac{kmm}{l^2} > 0$

$$\text{TAKE } \tilde{\theta}' = 0 \quad \tilde{\theta} = \tilde{\theta}' = 0 \Rightarrow \frac{1}{r} = C [1 + \epsilon \cos(0)] = C [1 + \epsilon]$$

MAX AT $\tilde{\theta} = 0$

$$\text{MINIMUM } r: \frac{1}{r_m} = C [1 + \epsilon]$$

ii) ATTRACTIVE CASE: $k > 0$ $C = \frac{km}{l^2} > 0$

TAKE $\tilde{\theta}' = 0$ $\tilde{\theta} = \tilde{\theta}' = 0 \Rightarrow \frac{1}{r} = C[1 + \epsilon \cos(0)] = C[1 + \epsilon]$

APSIDAL POINT IS ALSO AT $\tilde{\theta} = 0$

WE NOW NEED $\psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}(\frac{1}{r} \rightarrow 0)$

MAX AT $\tilde{\theta} = 0$

MINIMUM r : $\frac{1}{r_m} = C[1 + \epsilon]$

[IN THE ATTRACTIVE CASE]
 $\theta = 2\psi - \pi \Rightarrow \psi = \frac{\pi + \theta}{2}$

$$0 = 1 + \epsilon \cos \psi \Leftrightarrow -\frac{1}{\epsilon} = \cos \psi = \cos\left(\frac{\pi + \theta}{2}\right) = \underbrace{\cos \frac{\pi}{2}}_0 \cos \frac{\theta}{2} - \underbrace{\sin \frac{\pi}{2}}_1 \sin \frac{\theta}{2} = -\sin \frac{\theta}{2}$$

We now combine both cases:

$$\cot^2 \frac{\theta}{2} = \frac{1}{\tan^2 \frac{\theta}{2}} = \frac{\cos^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \frac{1 - \sin^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \frac{1}{\sin^2 \frac{\theta}{2}} - 1 = \epsilon^2 - 1 = \left(\frac{2sE}{ZZ'e^2} \right)^2 \Rightarrow \boxed{s = s(\theta, \epsilon) = \left| \frac{ZZ'e^2}{2E} \right| \cot \frac{\theta}{2}}$$

$$\frac{ds}{d\theta} = \left| \frac{ZZ'e^2}{2E} \right| \frac{d}{d\theta} \left(\frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} \right) = \left| \frac{ZZ'e^2}{2E} \right| \left(\frac{-\sin \frac{\theta}{2}}{2 \sin^2 \frac{\theta}{2}} - \frac{1}{2} \frac{\cos \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} \right) = - \left| \frac{ZZ'e^2}{4E} \right| \frac{1}{\sin^2 \frac{\theta}{2}} \quad \sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{d\mathbf{s}}{d\theta} \right| = \frac{1}{2} \left(\frac{ZZ'e^2}{2E} \right)^2 \frac{1}{\sin \frac{\theta}{2} \cos \frac{\theta}{2}} \frac{1}{\sin \frac{\theta}{2}}$$

$$\boxed{\sigma(\theta) = \frac{1}{4} \left(\frac{ZZ'e^2}{2E} \right)^2 \csc^4 \left(\frac{\theta}{2} \right)}$$

Rutherford scattering cross section
(for Coulomb force)

Remarks:

Remarks:

- 1) The cross section is the same for the repulsive and attractive cases, since it is proportional to $(zz')^2$.
- 2) It diverges like E^{-2} at low energies.
- 3) It diverges like θ^{-4} (VERY STRONG!!!) for small angles θ .
- 4) This classical result is exactly the same as the result obtained in Quantum Mechanics.

LITERATURE

4) This classical result is exactly the same as the result obtained in Quantum Mechanics.

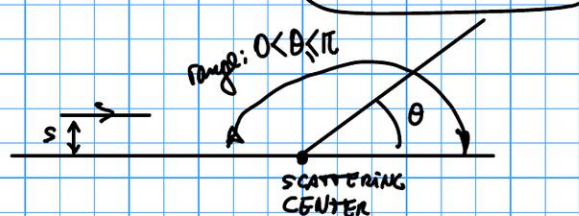
Total scattering cross section:

$$\sigma_T = \int \sigma(\Omega) d\Omega$$

In this case $\sigma(\theta) = \text{const} \frac{1}{\sin^4(\frac{\theta}{2})}$

$$d\Omega = 2\pi \sin\theta d\theta = 4\pi \sin\frac{\theta}{2} \cos\frac{\theta}{2} d\theta$$

$$\sigma_T = \frac{4\pi}{4} \left(\frac{ZZ'e^2}{2E} \right)^2 \int_0^\pi \frac{\sin\frac{\theta}{2} \cos\frac{\theta}{2}}{\sin^4\frac{\theta}{2}} d\theta$$



NOTATION:

Goldstein

Others

$\sigma(\Omega) \leftrightarrow \frac{d\sigma}{d\Omega}$ DIFFERENTIAL SCATTERING CROSS SECTION

$\sigma_T \leftrightarrow \sigma$ TOTAL SCATTERING CROSS SECTION

$$\sim \int_0^\pi \frac{d\theta}{(\theta/2)^3} = 8 \int_0^\pi \frac{d\theta}{\theta^3} = 8 \left. \frac{\theta^{-2}}{(-2)} \right|_0^\pi = +\infty \quad \text{STRONGLY DIVERGENT!}$$

CLASSICAL

$\sigma_T = +\infty$ for any long range potential

QUANTUM

$\sigma_T = +\infty$ for some long range potentials $V(r)$ do not decay fast enough at $r \rightarrow \infty$

CAVEAT

1) IN A DENSE CONDUCTING MATERIAL:

$$V = -\frac{k}{r} \longrightarrow V^{\text{SCREEN}}(r) = -\frac{k}{r} e^{-r/\lambda} \quad \text{SCREENED COULOMB POTENTIAL ("YUKAWA")}$$

(In non-conducting materials $V^{\text{SCREEN}}(r)$ may take a different form, but still will vanish much faster than $1/r$)